Functional Requirements

ID	Requirement	Module
FR-1	User can upload a food image (JPG, PNG or WEBP)	Food Recognition
FR-2	User can add optional voice (hold to record) or text context	Food Recognition
FR-3	System returns dish name, confidence, serving size (g), calories with range, protein, carbs, fat and a health score	Food Recognition
FR-4	System rejects unsupported file types and oversized files with a clear message	Food Recognition
FR-5	User can submit food for a freshness check and gets a result	Freshness Detector
FR-6	User can view a hygiene checklist, tick the items followed and get a hygiene score	Hygiene Monitor
FR-7	User can add, list and delete kitchen items with expiry dates, and request a recipe from a dish name or ingredients	Smart Kitchen
FR-8	User enters age, gender, height, weight, activity level, goal, diet type, cuisine, allergies, medical conditions and recent food history	Meal Planner
FR-9	System generates a personalized meal plan that avoids the listed allergies	Meal Planner
FR-12	User can upload an audio file and get a transcript	Voice
FR-10	System shows a readable error if the AI service fails	All
FR-11	Sidebar navigation between modules and an online/offline status indicator	UI

Non-functional requirements

- Response time: image analysis should return within a few seconds on a normal laptop.
- Input validation: age 12-100, height 100-230 cm, weight 30-250 kg in the meal planner.
- Privacy: health inputs are processed locally.
- Usability: single-page dark UI, works on desktop browsers.